

Determining P-Values in Experimental Data

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August 15, 2026

Abstract

The p-value is one of the most widely used quantities in statistical analysis. It provides a measure of how compatible an observed result is with a specified null hypothesis. However, a p-value is not a direct measure of the probability that a hypothesis is true. Instead, it is the probability of obtaining a result at least as extreme as the observed result, assuming that the null hypothesis is true.

This assignment explains how p-values are determined for different types of experimental data. Examples involving continuous, paired, categorical, and multi-group data are presented. The mathematical calculation of test statistics and their corresponding p-values is described for common statistical tests, including the t-test, analysis of variance (ANOVA), chi-square test, correlation, and linear regression. The importance of effect sizes, confidence intervals, assumptions, multiple testing, and appropriate interpretation of p-values is also discussed.

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1 Introduction

Scientific experiments frequently involve comparing groups, measuring associations, or determining whether an observed difference could reasonably have occurred by chance.

For example, a researcher may investigate whether a new treatment changes the concentration of an inflammatory biomarker compared with a control group. The observed means may differ, but the researcher must determine whether the difference is sufficiently large relative to the variability in the data.

Statistical hypothesis testing provides a framework for making this assessment.

The p-value is an important component of this framework. It is calculated from a statistical test and quantifies the extremeness of the observed data under the assumption that the null hypothesis is true.

A general hypothesis-testing framework consists of the following steps:

1. Define the research question.
2. State the null hypothesis (H_0).
3. State the alternative hypothesis (H_1).
4. Select an appropriate statistical test.
5. Calculate the test statistic.
6. Determine the p-value.
7. Compare the p-value with a predefined significance level.
8. Interpret the result together with the effect size and confidence interval.

2 Null and Alternative Hypotheses

The null hypothesis represents the default assumption that there is no effect, difference, or association.

For example, suppose an experiment compares a treatment group with a control group. The hypotheses may be written as

$$H_0 : \mu_T = \mu_C$$

and

$$H_1 : \mu_T \neq \mu_C$$

where μ_T is the population mean for the treatment group and μ_C is the population mean for the control group.

The null hypothesis states that the population means are equal, while the alternative hypothesis states that they are different.

The alternative hypothesis determines whether the test is one-sided or two-sided.

For a two-sided test:

$$H_1 : \mu_T \neq \mu_C$$

For a one-sided test:

$$H_1 : \mu_T > \mu_C$$

or

$$H_1 : \mu_T < \mu_C$$

The direction of the hypothesis should normally be determined before examining the experimental results.

3 What Is a P-Value?

A p-value can be formally defined as the probability, assuming that the null hypothesis is true, of observing a test statistic at least as extreme as the value obtained from the experiment.

In general,

$$p = P(\text{test statistic at least as extreme as observed} \mid H_0)$$

This definition contains an important condition:

The calculation assumes that H_0 is true.

Therefore, a p-value is **not** the probability that the null hypothesis is true.

For example, if

$$p = 0.03,$$

this does not mean that there is a 3% probability that H_0 is true.

Instead, it means that if H_0 were true, the probability of obtaining a result at least as extreme as the observed result would be approximately 3%.

4 Statistical Significance

A significance level, commonly represented by α , is selected before the analysis.

A frequently used value is

$$\alpha = 0.05.$$

The decision rule is

$$p < \alpha \Rightarrow \text{Reject } H_0$$

and

$$p \geq \alpha \Rightarrow \text{Fail to reject } H_0.$$

For example, if

$$p = 0.021$$

and

$$\alpha = 0.05,$$

then

$$0.021 < 0.05.$$

Therefore, the null hypothesis would be rejected.

However, statistical significance does not necessarily mean that the observed effect is scientifically or clinically important. The magnitude of the effect should also be considered.

5 Statistical Hypotheses

Hypothesis testing begins by translating a scientific research question into statistical hypotheses. The two fundamental hypotheses are the null hypothesis and the alternative hypothesis.

5.1 Null Hypothesis

The null hypothesis, denoted by H_0 , represents the default assumption that there is no statistically detectable effect, difference, or association.

For example, suppose a researcher wants to determine whether a treatment changes the mean concentration of a biomarker compared with a control group.

The null hypothesis can be written as:

$$H_0 : \mu_T = \mu_C$$

This means that the population mean in the treatment group is equal to the population mean in the control group.

The null hypothesis does not necessarily mean that the groups have exactly identical observations. Rather, it represents the assumption about the population parameter being tested.

5.2 Alternative Hypothesis

The alternative hypothesis, denoted by H_1 or H_A , represents the effect, difference, or association that the researcher is investigating.

For a two-sided test:

$$H_1 : \mu_T \neq \mu_C$$

This asks whether the two population means are different in either direction.

A one-sided alternative hypothesis can also be specified.

For example:

$$H_1 : \mu_T > \mu_C$$

tests whether the treatment mean is greater than the control mean.

Alternatively:

$$H_1 : \mu_T < \mu_C$$

tests whether the treatment mean is smaller than the control mean.

The choice between a one-sided and two-sided hypothesis should be based on the scientific question and experimental design and should normally be specified before examining the results.

6 Types of Statistical Errors

Hypothesis testing involves making a decision about the null hypothesis. Because decisions are made using sample data rather than the entire population, statistical errors are possible.

There are two fundamental types of errors.

6.1 Type I Error

A Type I error occurs when the null hypothesis is rejected even though the null hypothesis is actually true.

In simple terms, this is a:

False positive

For example, a statistical analysis may conclude that a treatment has an effect when, in reality, there is no effect in the population.

The probability of making a Type I error is controlled by the significance level α .

Thus:

$$P(\text{Type I error}) = \alpha$$

under the assumptions of the statistical test.

If

$$\alpha = 0.05,$$

the testing procedure is designed to have a 5% Type I error rate in the long run when the null hypothesis is true.

6.2 Type II Error

A Type II error occurs when the null hypothesis is not rejected even though a real effect exists.

In simple terms, this is a:

False negative

The probability of a Type II error is commonly represented by β :

$$P(\text{Type II error}) = \beta.$$

The statistical power of a test is defined as:

$$\text{Power} = 1 - \beta.$$

Power therefore represents the probability of detecting an effect when that effect truly exists, under the specified assumptions.

Table 1: Decisions in hypothesis testing

	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

7 Choosing the Pre-Specified Significance Level

The significance level is represented by α and should normally be chosen **before examining the experimental results**.

It represents the maximum Type I error rate that the researcher is willing to tolerate under the statistical testing framework.

A commonly used significance level is:

$$\alpha = 0.05.$$

However, 0.05 is not a universal requirement and does not represent a mathematical law.

Other significance levels may be appropriate depending on the research question and consequences of statistical errors.

Common examples include:

$$\alpha = 0.10, \quad \alpha = 0.05, \quad \alpha = 0.01.$$

A smaller value of α provides a more stringent criterion for rejecting the null hypothesis.

For example:

$$\alpha = 0.01$$

requires stronger statistical evidence against H_0 than:

$$\alpha = 0.05.$$

The choice of α should be based on factors such as:

- the consequences of a false positive;
- the consequences of a false negative;
- the scientific field;
- the purpose of the study;

- the study design;
- whether the analysis is exploratory or confirmatory;
- and whether multiple hypotheses are being tested.

For example, in a situation where a false positive could have serious consequences, a more stringent significance level may be appropriate.

Conversely, in exploratory research, a less stringent threshold may sometimes be considered, provided that it is justified and clearly reported.

Most importantly, the significance level should not be changed after examining the p-value simply to make a result statistically significant.

For example, a researcher should not obtain:

$$p = 0.07$$

and then decide:

$$\alpha = 0.10$$

solely because this makes the result significant.

Such an approach changes the statistical decision rule after seeing the data and can increase the risk of false-positive conclusions.

8 Relationship Between α and the P-Value

The significance level and p-value have different roles.

The significance level is selected **before** the analysis:

$$\alpha = \text{pre-specified decision threshold}$$

The p-value is calculated **from the observed data**:

$$p = \text{evidence against } H_0 \text{ under the specified model}$$

The two quantities are then compared.

If

$$p < \alpha,$$

the null hypothesis is rejected.

If

$$p \geq \alpha,$$

the null hypothesis is not rejected.

For example, suppose:

$$\alpha = 0.05$$

was specified before the experiment and the analysis produces:

$$p = 0.024.$$

Because

$$0.024 < 0.05,$$

the result is statistically significant at the 5% level.

In contrast, if:

$$p = 0.08,$$

then:

$$0.08 > 0.05,$$

and the result would not be statistically significant at the 5% level.

A non-significant result should not automatically be interpreted as proof that no effect exists. It means that the evidence was insufficient to reject the null hypothesis under the specified statistical framework.

Similarly, statistical significance does not necessarily imply that an effect is large, clinically important, or biologically meaningful.

9 From Hypothesis to P-Value

The complete hypothesis-testing process can therefore be summarized as:

Research question → Hypotheses → Choose α → Collect data
 → Test statistic → Probability distribution → p -value → Statistical decision

For example, for a two-group experiment:

Two groups $\rightarrow t$ -test $\rightarrow t = 2.47 \rightarrow t$ -distribution $\rightarrow p = 0.024$.

The p-value is therefore not determined by the value 0.05.

Instead, 0.05 is a commonly selected significance threshold against which the calculated p-value is compared.

10 How Is a P-Value Actually Calculated?

The calculation of a p-value generally follows three major stages:

Data $\rightarrow$ Test statistic $\rightarrow$ Probability distribution $\rightarrow$ P-value.

For example, in a t-test, the data are used to calculate a t statistic. The observed t statistic is then compared with the theoretical t -distribution.

For a two-sided t-test,

$$p = 2P(T \geq |t_{\text{obs}}|)$$

where T follows the appropriate t-distribution.

Different statistical tests use different probability distributions.

Table 2: Examples of statistical tests and their test statistics

Data/Question	Test	Statistic
One mean	One-sample t-test	t
Two independent means	Independent t-test	t
Paired measurements	Paired t-test	t
Three or more means	ANOVA	F
Categorical variables	Chi-square test	χ^2
Linear association	Pearson correlation	r
Prediction	Linear regression	t or F
Non-normal two-group data	Mann–Whitney U	U

11 Example 1: One-Sample T-Test

Suppose a laboratory has a reference mean concentration of

$$\mu_0 = 100 \text{ pg/mL}.$$

A researcher measures the concentration in 10 experimental samples and obtains

$$\bar{x} = 108 \text{ pg/mL}$$

with a sample standard deviation of

$$s = 10 \text{ pg/mL.}$$

The hypotheses are

$$H_0 : \mu = 100$$

and

$$H_1 : \mu \neq 100.$$

The one-sample t-statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Substituting the values gives

$$t = \frac{108 - 100}{10/\sqrt{10}}.$$

Therefore,

$$t \approx 2.53.$$

The degrees of freedom are

$$df = n - 1 = 9.$$

The two-sided p-value is therefore obtained from the t-distribution with 9 degrees of freedom:

$$p = 2P(T_9 \geq 2.53).$$

This gives approximately

$$p \approx 0.032.$$

Because

$$0.032 < 0.05,$$

the null hypothesis is rejected.

The result provides statistical evidence that the population mean differs from 100 pg/mL.

12 Example 2: Independent Two-Sample T-Test

Consider an experiment comparing a control group with a treatment group.

Suppose the results are:

Table 3: Example of continuous experimental data

Group	Sample size	Mean	SD
Control	10	500	80
Treatment	10	420	70

The research question is whether the treatment changes the measured variable.

The hypotheses are

$$H_0 : \mu_T = \mu_C$$

and

$$H_1 : \mu_T \neq \mu_C.$$

For Welch's two-sample t-test, the test statistic is

$$t = \frac{\bar{x}_T - \bar{x}_C}{\sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}}.$$

Substituting the values:

$$t = \frac{420 - 500}{\sqrt{\frac{70^2}{10} + \frac{80^2}{10}}}.$$

Thus,

$$t = \frac{-80}{\sqrt{490 + 640}}$$

and

$$t \approx -2.38.$$

The corresponding degrees of freedom are approximately 17.5.

The two-sided p-value is then obtained from the t-distribution:

$$p = 2P(T_{17.5} \geq |-2.38|).$$

The p-value is approximately

$$p \approx 0.029.$$

Since

$$p < 0.05,$$

there is evidence of a statistically significant difference between the two groups.

13 Example 3: Paired Data

Paired data occur when the same experimental subjects are measured twice.

For example, a researcher may measure a biomarker before and after treatment in the same subjects.

Instead of analyzing the two measurements as independent groups, the difference for each participant is calculated:

$$d_i = X_{\text{after},i} - X_{\text{before},i}.$$

The mean difference is

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i.$$

The paired t-statistic is

$$t = \frac{\bar{d}}{s_d / \sqrt{n}},$$

where s_d is the standard deviation of the paired differences.

The hypotheses are

$$H_0 : \mu_d = 0$$

and

$$H_1 : \mu_d \neq 0.$$

The p-value is then calculated using the t-distribution with

$$df = n - 1.$$

The important point is that the analysis focuses on the differences within subjects rather than treating the observations as independent.

14 Example 4: Analysis of Variance

When more than two independent groups are compared, performing multiple t-tests increases the probability of obtaining a false positive.

Analysis of variance (ANOVA) provides a method for testing whether there is evidence that at least one group mean differs from the others.

Suppose an experiment contains four treatment groups:

$$G_1, \quad G_2, \quad G_3, \quad G_4.$$

The null hypothesis is

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4.$$

The alternative hypothesis is

$$H_1 : \text{At least one population mean is different.}$$

ANOVA separates the total variability into between-group and within-group components.

The F-statistic is

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}.$$

The between-group mean square is

$$MS_{\text{between}} = \frac{SS_{\text{between}}}{k - 1},$$

while the within-group mean square is

$$MS_{\text{within}} = \frac{SS_{\text{within}}}{N - k}.$$

Therefore,

$$F = \frac{SS_{\text{between}}/(k - 1)}{SS_{\text{within}}/(N - k)}.$$

The p-value is obtained from the F-distribution:

$$p = P(F_{k-1, N-k} \geq F_{\text{obs}}).$$

For example, suppose an ANOVA produces

$$F(3, 36) = 6.42$$

with

$$p = 0.0015.$$

Because

$$p < 0.05,$$

the overall null hypothesis is rejected.

However, ANOVA alone does not identify which groups differ. A post-hoc procedure such as Tukey's honestly significant difference test may then be used to investigate pairwise differences.

15 Example 5: Chi-Square Test for Categorical Data

Not all experimental data are continuous.

Suppose researchers investigate whether treatment status is associated with disease status.

The observations could be summarized as follows:

Table 4: Hypothetical categorical data

	Disease	No disease
Treatment	10	40
Control	25	25

The null hypothesis is

$$H_0 : \text{Treatment and disease status are independent.}$$

The chi-square statistic is

$$\chi^2 = \sum \frac{(O - E)^2}{E},$$

where

O = observed frequency

and

E = expected frequency.

Expected frequencies are calculated as

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}}.$$

After calculating the chi-square statistic, the p-value is obtained from the chi-square distribution with the appropriate degrees of freedom.

For a contingency table,

$$df = (r - 1)(c - 1),$$

where r is the number of rows and c is the number of columns.

For a 2×2 table,

$$df = (2 - 1)(2 - 1) = 1.$$

16 Example 6: Pearson Correlation

A researcher may wish to determine whether two continuous variables are associated.

For example, suppose the concentration of a biomarker is compared with a physiological measurement.

Pearson's correlation coefficient is

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}.$$

The correlation coefficient ranges from

$$-1 \leq r \leq 1.$$

A positive value indicates a positive linear association, while a negative value indicates a negative linear association.

To test the null hypothesis

$$H_0 : \rho = 0,$$

the following statistic can be used:

$$t = r \sqrt{\frac{n - 2}{1 - r^2}}.$$

The statistic follows a t-distribution with

$$df = n - 2.$$

The resulting p-value tests whether there is evidence of a non-zero linear association.

17 Example 7: Linear Regression

Regression is often used when the objective is to determine whether one or more predictor variables are associated with an outcome.

A simple linear regression model can be written as

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i.$$

Here:

- Y_i is the outcome variable,
- X_i is the predictor,
- β_0 is the intercept,
- β_1 is the regression coefficient,
- ϵ_i is the error term.

A common hypothesis is

$$H_0 : \beta_1 = 0$$

versus

$$H_1 : \beta_1 \neq 0.$$

The estimated coefficient can be tested using

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}.$$

The p-value is obtained from the corresponding t-distribution.

For example, suppose a regression analysis produces

$$\hat{\beta}_1 = -2.50$$

with

$$SE(\hat{\beta}_1) = 0.80.$$

Then

$$t = \frac{-2.50}{0.80} = -3.125.$$

The corresponding p-value can then be calculated using the t-distribution with the appropriate degrees of freedom.

18 One-Sided and Two-Sided P-Values

The definition of the p-value depends on the alternative hypothesis.

For a two-sided test,

$$H_1 : \theta \neq \theta_0,$$

evidence in either direction is considered extreme.

The p-value is therefore

$$p = 2P(T \geq |t_{\text{obs}}|).$$

For a right-sided test,

$$H_1 : \theta > \theta_0,$$

the p-value is

$$p = P(T \geq t_{\text{obs}}).$$

For a left-sided test,

$$H_1 : \theta < \theta_0,$$

the p-value is

$$p = P(T \leq t_{\text{obs}}).$$

The choice between one-sided and two-sided testing should be based on the research question and should not be changed simply because the observed data are in a particular direction.

19 Confidence Intervals and P-Values

P-values should not be interpreted in isolation.

A confidence interval provides information about the plausible magnitude of an effect.

For example, suppose a treatment reduces a biomarker concentration by

$$\Delta = -80 \text{ pg/mL}$$

with a 95% confidence interval of

$$[-145, -15] \text{ pg/mL.}$$

The interval does not contain zero, which is consistent with a two-sided test producing

$$p < 0.05.$$

Confidence intervals are particularly useful because they provide information about both the estimated effect and its uncertainty.

20 Effect Size Versus Statistical Significance

A statistically significant result is not necessarily a large or scientifically important effect.

For example, with a very large sample size, a very small difference can produce

$$p < 0.001.$$

Conversely, a biologically meaningful effect may fail to reach statistical significance when the sample size is small or the data are highly variable.

Therefore, experimental reports should ideally include:

- p-values,
- effect sizes,
- confidence intervals,
- sample sizes,
- measures of variability,
- and the statistical method used.

21 Multiple Comparisons

When many statistical tests are performed, the probability of obtaining at least one statistically significant result by chance increases.

For example, if 20 independent tests are performed using

$$\alpha = 0.05,$$

the probability of obtaining at least one false positive is

$$1 - (1 - \alpha)^{20}.$$

Therefore,

$$1 - (0.95)^{20} \approx 0.642.$$

This means that under idealized independence assumptions, there is approximately a 64.2% chance of obtaining at least one nominally significant result when all null hypotheses are actually true.

Methods for controlling multiple comparisons include:

- Bonferroni correction,
- Holm correction,
- Tukey's HSD,
- Benjamini–Hochberg false discovery rate control.

The appropriate correction depends on the research design and scientific objective.

22 P-Value Interpretation in Biological Experiments

In biological and biomedical research, p-values are frequently used to compare experimental groups.

For example, suppose a researcher investigates whether a treatment reduces TNF- α concentration.

The hypotheses could be

$$H_0 : \mu_{\text{treatment}} = \mu_{\text{control}}$$

and

$$H_1 : \mu_{\text{treatment}} \neq \mu_{\text{control}}.$$

Suppose the analysis produces

$$F(3, 28) = 14.8$$

with

$$p < 0.001.$$

The appropriate interpretation is:

There is strong statistical evidence that mean TNF- α concentrations differ among the treatment groups.

However, this result alone does not demonstrate which groups differ. Post-hoc comparisons are required to identify specific pairwise differences.

Furthermore, statistical significance does not establish biological importance. The magnitude of the reduction, confidence intervals, experimental design, sample size, and biological mechanism should also be considered.

23 Common Misinterpretations of P-Values

Several incorrect interpretations of p-values are common.

23.1 A p-value is not the probability that H_0 is true

Incorrect:

$$p = 0.03 \Rightarrow P(H_0) = 0.03.$$

This conclusion is not justified by a conventional frequentist p-value.

23.2 A small p-value does not measure effect size

A p-value indicates evidence against the null hypothesis under the model. It does not indicate how large the effect is.

23.3 A non-significant result does not prove no effect

If

$$p > 0.05,$$

the appropriate conclusion is generally that there is insufficient evidence to reject the null hypothesis.

It is not necessarily correct to conclude:

“There is no effect.”

24 Choosing the Appropriate Statistical Test

The statistical test should be selected based on the experimental design, data type, number of groups, independence of observations, and assumptions of the statistical model.

Table 5: General guide for selecting statistical tests

Research Question	Data Structure	Possible Test
Compare one mean with a reference	One continuous sample	One-sample t-test
Compare two independent means	Two independent continuous groups	Independent t-test
Compare before and after measurements	Paired continuous observations	Paired t-test
Compare three or more means	Multiple independent groups	ANOVA
Compare categorical proportions	Categorical variables	Chi-square/Fisher's exact test
Measure linear association	Two continuous variables	Pearson correlation
Predict a continuous outcome	Continuous outcome and predictors	Linear regression
Compare non-normal independent groups	Ordinal/non-normal continuous data	Mann-Whitney U

The table provides general guidance rather than a universal decision rule. Statistical assumptions should always be evaluated.

25 Using Statistical Software

In practice, p-values are usually calculated using statistical software rather than manually evaluating probability distributions.

For example, in R, a two-sample t-test can be performed using:

```
control <- c(500, 520, 480, 510, 490)
treatment <- c(420, 440, 400, 430, 410)
```

```
t.test(control, treatment)
```

For an ANOVA:

```
model <- aov(value ~ group, data = dataset)
summary(model)
```

For a chi-square test:

```
chisq.test(table(dataset$treatment,
                  dataset$disease))
```

For correlation:

```
cor.test(dataset$x, dataset$y,  
         method = "pearson")
```

For linear regression:

```
model <- lm(y ~ x, data = dataset)  
summary(model)
```

The software calculates the test statistic and evaluates its probability under the relevant statistical distribution.

However, software does not determine whether the test is scientifically appropriate. That decision must be made by the researcher.

26 General Mathematical Framework

Many hypothesis tests can be represented using the general framework

$$\text{Test statistic} = \frac{\text{Observed effect} - \text{Null expected effect}}{\text{Standard error of the effect}}.$$

This can be expressed conceptually as

$$T = \frac{\text{Signal}}{\text{Noise}}.$$

A large absolute value of the test statistic means that the observed effect is large relative to its expected sampling variability.

The p-value is then obtained by evaluating the probability distribution of the test statistic under H_0 .

Thus,

$\text{Data} \rightarrow \text{Effect} \rightarrow \text{Standard Error} \rightarrow \text{Test Statistic} \rightarrow \text{P-Value}$

This framework explains why different statistical tests produce different test statistics but ultimately use the same fundamental logic.

27 Conclusion

The p-value is a measure of how unusual the observed experimental result would be if the null hypothesis were true. It is calculated by first selecting an appropriate statistical test, calculating a test statistic, and then determining the probability of obtaining a value at least as extreme as the observed statistic under the null hypothesis.

Different experimental designs require different statistical procedures. Continuous measurements may be analyzed using t-tests, ANOVA, correlation, or regression, while categorical observations may require chi-square or Fisher’s exact tests. Non-parametric procedures can be used when the assumptions of parametric methods are not appropriate.

A p-value should not be interpreted alone. Effect size, confidence intervals, sample size, variability, biological relevance, assumptions, and multiple-testing considerations are all important when interpreting experimental results.

The fundamental principle can therefore be summarized as

$$p = P(\text{data at least as extreme as observed} \mid H_0)$$

and the final statistical conclusion should always be connected to the scientific question being investigated.

References

1. Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver and Boyd.
2. Student. (1908). The probable error of a mean. *Biometrika*, 6(1), 1–25.
3. Montgomery, D. C. (2019). *Design and Analysis of Experiments*. John Wiley & Sons.
4. Field, A. (2018). *Discovering Statistics Using IBM SPSS Statistics*. SAGE Publications.
5. Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2), 129–133.